Week 2: Introduction to Pre-analysis Plans

Motivation and Important Components

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Lecture Outline

1. Motivation for pre-analysis plans
2. Links to pre-registration
3. Key components
    1. Introduction
    2. Experimental Design
    3. Analysis
    4. Implementation
4. Example of a good pre-analysis plan
5. Example of a bad pre-analysis plan

Motivation for Pre-analysis Plans

The current credibility issues in science

ioannidis_why_2005 paper claims over 50% of current findings are false. Many agree the number is overblown, but the criticisms of false positives and the general structure of the research environment are not.

P-hacking: Term used to describe the practice of manipulating results of statistical tests to find statistically significant results. Also known as data dredging.

When was this issue first described?

Motivation for Pre-analysis Plans

I don't know, but they have unfortunately been around for a while. They were previously described in 1967, along with several other issues, as seen in **lakens_which_2024**.

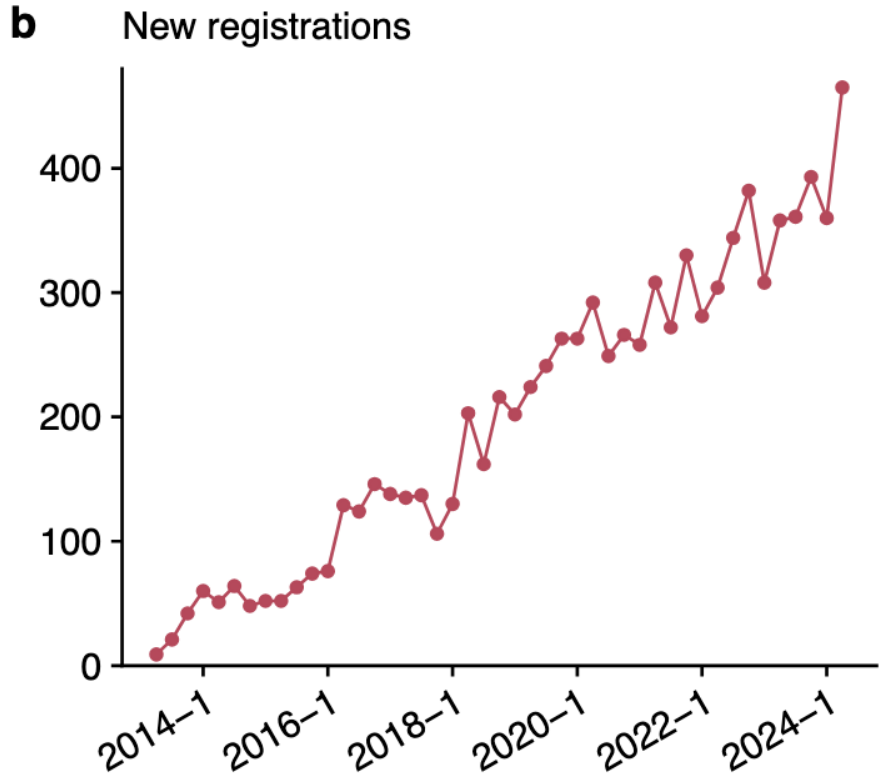
Economics not immune, adopts and develops some counter-measures

necker_scientific_2014 highlights that several questionable research practices in economics persist, but many sub-fields and journals within economics develop countermeasures.

Motivation for Pre-analysis Plans

Pre-registration and pre-analysis plans are designed to directly outline the experiment and analysis performed such that deviations from that plan have to be limited and extensively justified. Pre-registration tends to be a bit less detailed on experimental design, but includes extra elements such as ethics.

For more opinions on the effectiveness and current state of these measures, see **ofosu_pre-analysis_2023** and **coffman_pre-analysis_2015**.



Rising number of studies registered on AEA RCT registry per quarter (**imai_pre-registration_2025**).

Structure of a Pre-analysis Plan

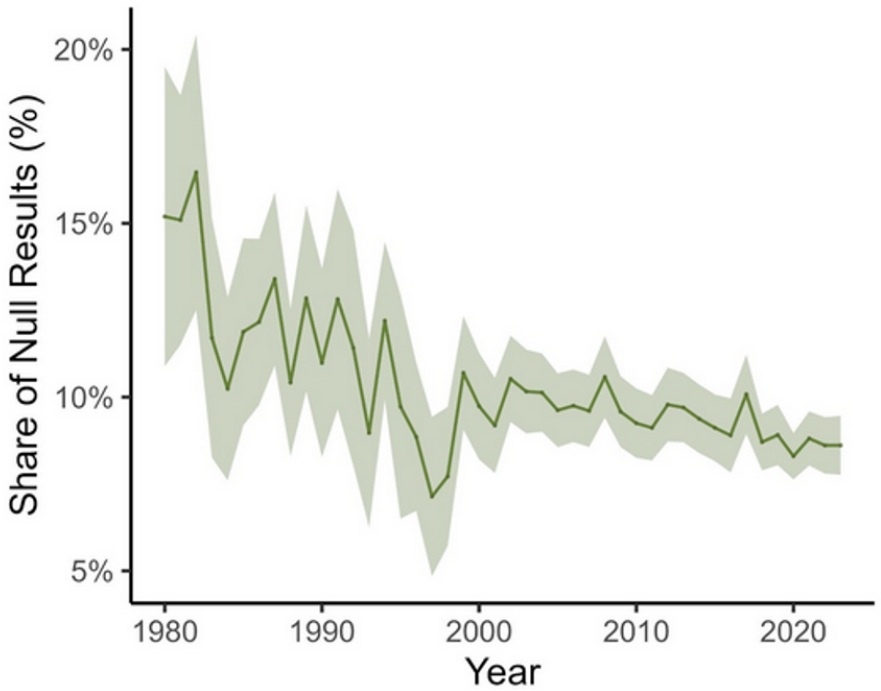
Note: We will be focusing on the structures for pre-analysis plans in the fields of development, behavioural, and experimental economics as found on the AEA RCT Registry. Other social sciences, particularly psychology, also use these techniques. *The sections discussed are generally included, but are not entirely fixed.*

1. **Introduction**: Describes the motivation of the experiment and connects it to the literature. Similar to an introduction in standard scientific papers.
2. **Experimental Design**: Outlines sampling domain and procedure, as well as treatment structure.
3. **Analysis**: Outlines the *hypothesis to be tested in the main analysis*, the outcomes, and how the experimental design will accommodate this (i.e. sufficient power).
4. **Implementation**: Describes the logistics of how the study will take place and potential issues that could impact study viability.

A Good Pre-analysis Plan: Hoces de la Guardia

Pre-registration, Reporting Guidelines and Publication Patterns in Economics fernando_hoces_de_la_guardia_pre-registration_2025

- Understand how scientists create and report pre-sprecified hypotheses
- A better estimate of publication bias
- A standardized way to find hypotheses and connect them to results
- Develop and test a method to encourage people to public all hypotheses



Decline Over Time: The reporting of null results has decreased from approximately 15% in 1980 to around 8.6% in 2023.

garg_causal_2025

Pre-analysis Plans: Introduction

The introduction focuses on outlining the **research question**, focusing on what **motivates** them to target this particular research question and what they **aim** to do by doing so.

Motivation focuses on informing the reader of the current state of the subject matter and literature on subject matter. Typically, a gap in the literature would be identified here.

Aim focuses on what outputs will come from the research. This includes quantitative estimates for values of interest and the practical value of the intervention, if relevant.

Introduction: Hoces de la Guardia

What would motivate Hoces de la Guardia et al. to look at pre-registration and how to improve its use?

What are the goals of the research team?

Introduction: Hoces de la Guardia

Answer: Publication bias is a major concern without good estimates of its rate of occurrence. By providing a follow-up intervention, they could help authors understand and remedy publication bias in their own work.

Goals of the research team:

- **Estimate of QRP:** A better estimate of publication bias
- **Connect pre-registration with analysis afterwards**: A standardized way to find hypotheses and connect them to results
- **Intervention to make pre-registration follow through more common**: Develop and test a method to encourage people to public all hypotheses

Experimental Design - Sampling and Population

Population - Understanding the research domain

Clearly delineate the population of the observational units, including theoretical justification for exclusion of certain units from the population. If there are multiple populations, they should be connected to each other or the research question.

Sample - finding an appropriate subpopulaiton for analysis

With a clearly defined population, identify characteristics for which units from the population(s) you will sample.

Population and Sampling: Hoces de la Guardia

Two relevant populations

- Pre-registrations/studies found on the AEA RCT registry
- Follow-up material from those registrations containing results of experiments concerning the hypotheses

Sampling Procedure: In this paper, the authors select a time range from 2015-2017 as their sample of pre-registrations. A *contingency* is included to state that they could also include studies from 2018. This brings them up to 1402 initial pre-registrations.

Experimental Design: Treatment and Randomization

Treatments and Interventions

Each treatment arm is described, both in terms of how it functions in the real world and the mechanism of change is is supposed to use to induce change on each unit it is applied to.

If the intervention is a new program designed by the research group, it must be described clearly.

Randomization

Randomization should be explained such that we can see how potential causal issues are addressed. Additional information about the population should be described if used in the randomization process.

Treatment and Randomization: Hoces de la Guardia

Interventions

Outline the different arms of support for contacted researchers and explain how they differ and overlap.

- Control: Only contacted to request pre-analysis plan for encoding
- Information, encouragement, empty report result
- Information, encouragement, pre-filled report result
- Information, encouragement, pre-filled report result, resource support

Randomization between arms was stratified along:

- Whether the pre-registration includes a pre-analysis plan
- (A proxy for) the type of economics the study is covering
- Which member of the research team is encoding the study

Why should be concerned with something such as which research team member encoding the data?

Analysis: Outcomes

Definition of Outcomes

Clearly outline what your dependent and independent variables are, how you measure and prioritize them, and how this relates back to the problem at hand. Are there edge cases in our example?

$$Y_{ki} = \frac{\sum_{h \in H_i} Y_{khi}}{N_{H_i}} \quad \text{for } k = 1, 2,$$

Model Specification

Consider whether you should consider effects as random or fixed, whether interactions are appropriate, or any transformations. How should heterogeneity be factored in? Refer to the literature!

$$Y_{it} = \mu_i + \delta_1 I_{t=1} + \delta_{1Z} (Z_i \times I_{t=1}) + \sum_{j=1}^{J} \tau_j (T_{ij} \times I_{t=1}) + \sum_{j=1}^{J} \gamma_j (Z_i \times T_{ij} \times I_{t=1}) + \epsilon_{it}, \quad t = 0, 1. \quad (3.3)$$

Designing Your Hypotheses

Confirmatory vs Exploratory

One of the main objectives of a pre-analysis plan is the clear specification of which hypotheses a research team aims to evaluate.

To improve salience of hypotheses, included hypotheses should

- Clearly be connected to the model specification
- Be classified by which type of outcome they concern (primary, secondary, heterogeneity)
- Have a clear null, alternative, and justification for the directionality of the test

Warning!

Each additional hypothesis you add will mean you have to account for it, or else you will have a multiple testing problem!

Analysis: Power

Power: The probability of detecting a false null hypothesis, given a specific alternative.

The power section of a pre-analysis plan should include

- Clear calculations for your each hypotheses
- A synthesis of from the calculations that indicate which hypotheses or outcome quantity is leading the contraint determining sample size
- An exploration of what theoretical elements could pose a risk to power (i.e. attrition)

Note: Generally, previous literature is referred to in the power calculations for getting estimates for expected effect sizes or standard deviations. Sometimes, the minimum detectable effect is expressed in standard deviations if that estimate is not available. This will be discussed further in the power lecture.

Implementation and Logistics

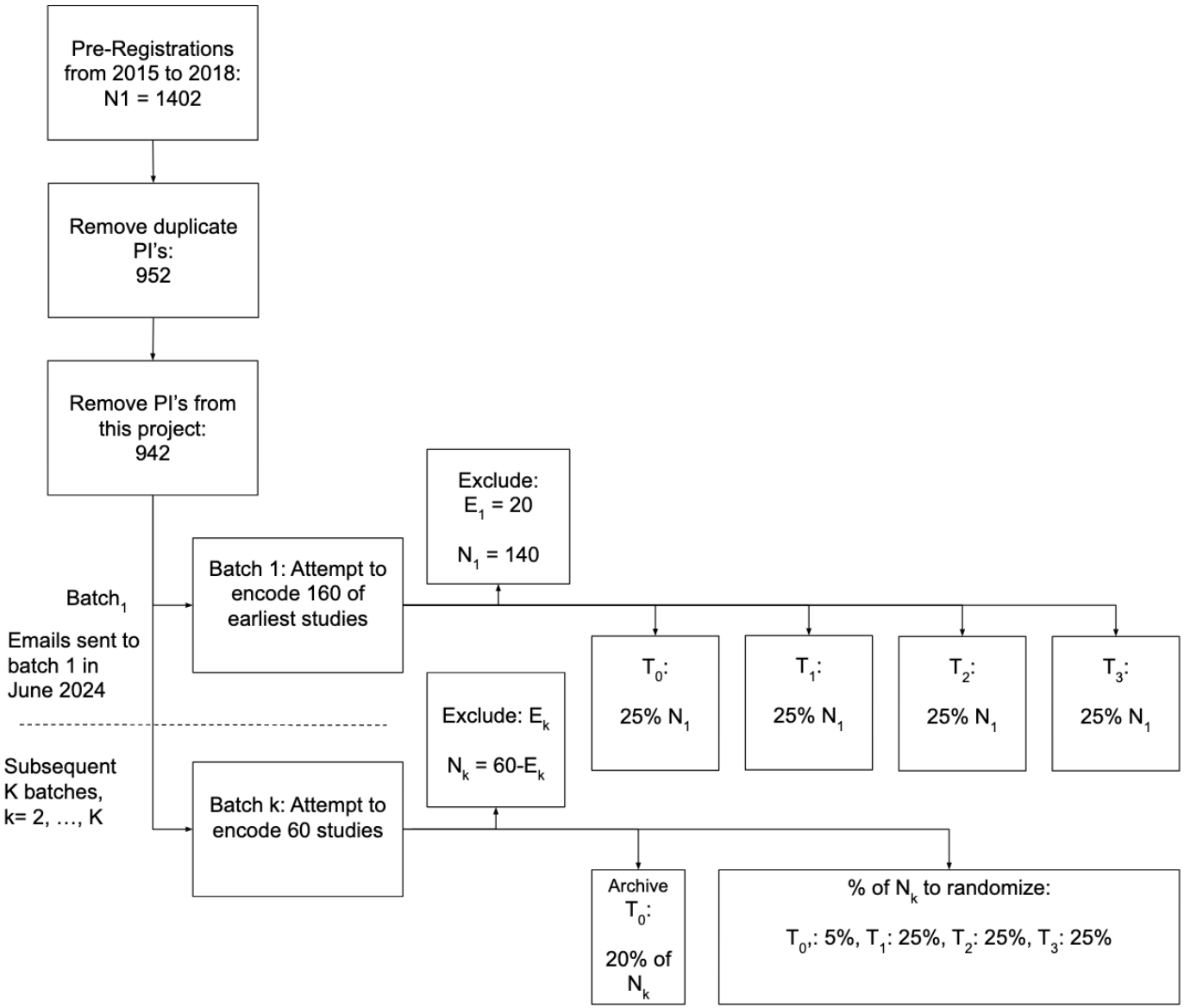
Dealing with the real world

This section is where research teams show they can get from the theoretical constructs outlined in the previous sections to the real-world realities faced in their domain and among the population of their research units.

How will data be collected from sampled units?
How will units be convinced to participate?
How should units be excluded and why?
Is the intervention easily understood to the population?

What do we do when something goes wrong?

Implementation Logistics: Hoces de la Guardia



Implementation Logistics: Hoces de la Guardia

This example is above and beyond what we ask you to include, but thinking about your sampling logistics in a decision tree framework can be useful for developing a written outline as well.

A particularly clever example from **fernando_hoces_de_la_guardia_pre-registration_2025** is to have a plan to collapse different interventions together based on lack of observational units, and this is justified in this section of their pre-analysis plan.

Additional Tips from resources including J-PAL
maya_duru_pre-analysis_nodate

- **Brevity**: Though pre-analysis plans used to reach upwards of 30 pages, keeping your descriptions and outlines brief is now standard to still permit some flexibility. We will aim for 10-15 pages in this course.
- **Admit uncertainty**: It is good practice to admit where you are uncertain, whether it be in model specification, potential logistical issues, or something else. Do not make strong promises without sufficient justification.
- **Clear structure over long explanations**: Do not format a pre-analysis plan the same way you would a paper. It will not lend itself to brevity. List formats are commonly built into each of the sections we have discussed.
- **Look for examples**: In this class and beyond, referencing (and citing) the literature is a good way to avoid going too far astray.

Citations and Resources I